

The following document is submitted by email to ostp-ai-rfi@nitrd.gov in response to the Request For Information (RFI) on the Development Of An Artificial Intelligence (AI) Action Plan ("Plan").

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Abstract

The development of advanced AI alone or in synergistic combination with AI-enabled humanoid technologies ("Advanced AI Systems") is inevitable and is recognized as critical to fundamental United States interests. As the RFI states: "The Trump Administration recognizes that with the right government policies, the United States can solidify its position as the leader in AI and secure a brighter future for all Americans."

In order to "secure a brighter future for all Americans," among the AI Action Plan's highest priorities should be recognizing that Advanced AI Systems represent a fundamental break from historical patterns of technological labor unemployment and reemployment. We should not be promising a bright future for all Americans unless we can identify and provide adequate full employment opportunities for those able and willing to work, or unless we are prepared to provide trillions of dollars in Universal Subsistence Support to the overwhelming majority of what would otherwise have been gainfully employed Americans.

The historical pattern of technological unemployment and reemployment will not apply to the widespread deployment of Advanced AI Systems because such systems represent a historically qualitatively different velocity and type of technological advancement—one capable of replacing human labor across nearly all sectors of the economy through comprehensive replication of human cognitive and physical capabilities. This qualitative difference is evidenced by our current inability to articulate concrete ways the overwhelming majority of humans would meaningfully participate in such an AI economy—not due to limited imagination, but because the technology's unprecedented defining characteristic is its potential to eliminate the necessity of human involvement. Unlike previous technological revolutions, where new roles for human labor were visible even in early stages, Advanced AI Systems leave no clear path for large scale human economic participation. In due course, the deployment of Advanced AI Systems will not render human labor more productive; rather, it will render human labor economically obsolete.

This document introduces three key frameworks: first, previous technological transitions have systematically misattributed machine productivity to human capability, and thus we have failed to appreciate the potential impact of AI systems on labor participation and unequal wealth distribution; second, faith in technological continuity has acquired quasi-religious characteristics that blind us to AI's Siren's Call; and third, there is no clear plan on how to dramatically enhance currently inadequate economic, taxation, and human engagement policies to secure, in the context of Advanced AI Systems, a minimally bright future for most Americans.

It should be appreciated that, as in other policy initiatives, in the context of advancing AI capabilities, President Trump administration's various well-intended actions, and in particular the work of DOGE, will act as a model and catalyst to accelerate labor displacement, not only in government but more widely, in private enterprises.

If, as many expect, the deployment of Advanced AI Systems is a game changer, then many of the rules will necessarily change.

This document briefly introduces the following topics:

- Breaking The Pattern Of Technological Employment
- Task-Specific AI And Immediate Labor Displacement
- The Historical Misattribution Of Productivity Gains
- Technological Progress As Religion
- AI's Siren Call
- Universal Subsistence Support
- Economic And Population Downward Spiral
- Reimagining Human Purpose
- About The Author

Breaking The Pattern Of Technological Employment

Prior to AGI/ASI level capabilities, dramatic labor displacement will begin to materialize with the introduction of Advanced AI Systems synergistically integrating Narrow AI or Artificial Narrow Intelligence (ANI). These systems include AI specialized for specific tasks or domains, such as task-specific Large Language Models (LLMs), Computer Vision Models specialized in visual perception and analysis, Expert AI Systems designed for specific professional domains, and advanced humanoid technologies incorporating vision, language, improved sensorimotor control, and enhanced dexterity and manipulation capabilities.

While in the early stages of Advanced AI Systems deployment, there will certainly be opportunities for a very limited number of specialists, these opportunities do not overcome the forthcoming widespread labor displacement. The opacity around future employment signals a genuine absence of necessary human economic roles rather than mere predictive uncertainty.

Unlike previous technological revolutions where new machinery and systems inherently created visible chains of human involvement in their production, maintenance, and operation, advanced AI systems will in due course eliminate the need for human participation across the entire value chain, including next-generation Advanced AI Systems' own advancement and reproduction.

During the Industrial Revolution, powered looms displaced hand weavers but created obvious new roles for machine operators, factory supervisors, and maintenance personnel. The rise of automobiles eliminated horse-related occupations but generated clear opportunities in manufacturing, mechanical repair, and supporting industries like gas stations and parts suppliers. Even the computer revolution, while eliminating many clerical positions, beyond manufacturing, spawned entire new fields in software development, IT support, and data analysis.

The key difference with Advanced AI Systems lies in their fundamental nature as widely implementable at a relatively low cost compared to the total costs associated with human employment. Unlike specialized technologies that augment or replace specific human tasks, Advanced AI Systems will be designed to match or exceed human abilities across all domains—physical, cognitive, and creative. This comprehensive capability makes Advanced AI Systems qualitatively different from all previous technological innovations. Past experiences of technological unemployment and reemployment are simply not analogous to the impact of Advanced AI Systems.

The Structural Impediments to Human-AI Collaborative Employment

Proposals that humans will find substantial employment through collaboration with AI systems fail to address four critical structural obstacles. First, the scale problem: even if long-term human-AI collaborative roles exist, they will likely represent an insignificant fraction of jobs eliminated, creating a massive employment deficit. Second, the capability ceiling dilemma: many proposed collaborative frameworks assume permanent human advantages in creativity, judgment, or emotional intelligence that advanced AI systems may eventually match or exceed. Third, the economic incentive reality: in competitive markets, the cost differential between human-AI collaboration versus fully automated solutions will continuously pressure organizations to eliminate human components from workflows. Fourth, the distribution challenge: the specialized skills required for remaining human roles in an AI economy may be inaccessible to most displaced workers due to aptitude, educational barriers, or geographic limitations, potentially creating a small class of employable specialists while leaving the majority without viable economic participation options.

Evidence of the Pattern Across Industries

Current professions thought to be relatively immune to automation, such as medical practitioners, are already seeing their capabilities matched or exceeded by AI systems across multiple domains. In surgery, the progression is particularly telling: first, robotic arms became the primary contact point with patients, with human surgeons controlling them remotely. The next step, already underway, is replacing human control with AI-driven systems using advanced visual recognition and precision control—eliminating the need for human surgical expertise entirely. This progression mirrors what we're likely to see across many industries—a gradual shift from AI/robots as tools to autonomous systems that completely replace human involvement.

Beyond the operating room, AI systems are being enhanced to match or exceed human capabilities in diagnosis and treatment planning. Even the supposedly uniquely human aspect of medical care—empathy and bedside manner—is being effectively replicated by AI systems that can craft more empathetic responses than human physicians.

In the legal profession, current language models already possess the capability to provide comprehensive legal services, with the primary barrier being LLM providers' current strategic choice to avoid potential legal liabilities rather than any technological limitation. The logical endpoint of this progression will be fully automated legal proceedings where AI systems represent both plaintiffs and defendants, presenting cases to an AI judge who could process the entire body of relevant law and precedent in

minutes rather than months. This is not a far-fetched scenario—it simply represents the natural optimization of legal processes once we remove the artificial constraint of human processing speeds and address the obstructions imposed by the profession's self-interests.

The creative industries, often cited as a refuge for human employment, are similarly vulnerable. The rapid advancement in AI-generated content, from writing to visual art to music composition, demonstrates that even these supposedly human-centric domains are not immune. The entertainment industry's response, as evidenced by recent SAG-AFTRA legislation regarding AI replicas of performers, exemplifies a broader pattern of professional resistance through organized labor. This resistance, while understandable, may ultimately prove futile against the economic imperatives driving AI adoption.

Indeed, the increasing activism of unions and professional associations across various sectors represents a significant but temporary barrier to AI displacement. These organizations are adopting increasingly aggressive strategies to preserve human labor—demanding contractual guarantees against AI replacement, advocating for regulatory restrictions on AI deployment, and seeking to establish protected domains of exclusively human work. However, these efforts face three fundamental challenges: first, they can only delay rather than prevent the eventual economic advantages of AI adoption; second, they risk accelerating their own obsolescence by driving companies to develop fully automated alternatives that circumvent human labor entirely; and third, a large pool of unemployed humans will undermine a union's bargaining position. The pattern is already visible in how companies respond to labor actions by investing more heavily in automation, effectively transforming temporary work stoppages into permanent labor elimination.

Task-Specific AI And Immediate Labor Displacement

Smaller AI companies are developing highly specialized, task-specific AI models targeting discrete job functions. These focused AI systems, while lacking general intelligence, can effectively replicate and replace specific human roles: a medical office receptionist handling appointments and insurance verification, a repair service AI managing scheduling and dispatch, or a customer service AI handling routine inquiries and complaints.

This pattern of targeted displacement is particularly powerful because it allows for incremental adoption without requiring massive infrastructure changes. Small software companies can develop these specialized models by fine-tuning existing language models and combining them with basic process automation. The investment required is relatively modest compared to developing general AI or robotics, making the business case compelling even for small and medium-sized businesses.

The advantages of AI voice-enabled customer service over current systems are substantial and immediate. Where traditional systems force customers through rigid menu trees, requiring them to listen to entire option lists and frequently repeat information, AI systems enable natural conversation where users simply state their problems in their own words. The AI can adaptively ask clarifying questions based on

responses, maintaining context throughout the interaction without requiring repetition at different stages.

The potential impact on government bureaucracies is especially compelling. Current government service interactions often involve long wait times, multiple transfers between departments, repetitive form-filling, and frequent confusion about proper procedures. An AI system will transform this experience by providing instant, accurate information about complex regulations, guiding citizens through required documentation, and ensuring consistent interpretation of rules across all interactions.

The scope of potential AI automation in government administrative roles is staggering. Approximately 2.7 million government employees across federal, state, and local levels are currently engaged in administrative, paper-processing, and customer service roles that could be substantially automated by AI systems. A conservative estimate suggests another 1-1.5 million government positions involve significant administrative components that could be automated, bringing the total potential impact to around 4 million positions, representing approximately 15% of total government employment.

The Historical Misattribution Of Productivity Gains

Arguments for the resiliency of human labor in the face of technological progress are often associated with the idea that human productivity increases with technology. This historical narrative of increasing human productivity represents a uniquely consequential misattribution in economic thought. While this misattribution exists on a spectrum rather than as an absolute, the vast majority of what we label as human productivity improvements actually represents the productive capacity of capital equipment itself.

Consider the modern farmer with a GPS-guided tractor versus their historical counterpart with an ox-drawn plow. While the contemporary farmer has indeed developed new skills in operating software and navigation systems, these skills do not account for the massive productivity differential. The predominant productivity increase derives purely from the productive capacity of the capital equipment itself.

The business language surrounding technological investment systematically obscures this reality. Return on investment calculations are invariably presented in terms of "productivity improvements" when they are actually calculating a much simpler equation: $(\text{Cost of Human Labor Eliminated}) - (\text{Cost of Machine} + \text{Maintenance})$. This calculated ambiguity serves both psychological and social functions, maintaining the fiction that technology enhances rather than replaces human labor. Even the seemingly more honest term "labor-saving device" softens the reality by framing job elimination as mere savings.

This pattern of linguistic misdirection pervades business terminology. "Efficiency gains" really means labor reduction. "Performance enhancement" means labor reduction. "Streamlined operations" means labor reduction. "Optimizing workforce output" means labor reduction. The marketing of AI systems particularly exemplifies this tendency—vendors claim their systems will "make customer service representatives 300% more productive" rather than stating the reality: "our system will eliminate 75% of your

customer service positions." This deliberate obscuring of labor reduction behind productivity language reinforces the broader pattern of misattributing capital productivity to human capability.

Modern manufacturing crystallizes this pattern. The claim that today's factory worker is "more productive" than their counterpart from fifty years ago primarily describes the output of increasingly sophisticated machinery and automation systems. While workers have developed new skills in machine operation and monitoring, these skills represent a diminishing percentage of the total productive output. It is not inconsequential that we have begun to speak of "enterprise productivity."

This spectrum of attribution varies across different technological implementations. In software development, for instance, modern programmers might legitimately claim a higher percentage of productivity gains through their sophisticated understanding of complex development environments and architectural patterns. Yet even here, the majority of productivity improvements stem from the capabilities of modern development tools, libraries, and increasingly, AI coding assistants.

This systematic misattribution has profound implications for understanding economic inequality. The growing disparity between returns to labor and returns to capital isn't simply a policy outcome or market failure—it accurately reflects where productive capacity actually resides. What we've labeled as "human productivity improvements" has actually been the progressive transfer of productive capacity from human labor to capital equipment. The owners of capital have reaped the benefits of this misattributed productivity, while human labor's fundamental contribution has remained relatively constant or even diminished.

The emergence of Advanced AI Systems represents not just another step in this progression but its logical conclusion—the final transfer of all productive capacity to capital and more specifically to Advanced AI Systems. The current state of human-AI interaction represents a transitional phase in this progression. Consider the development of this analysis itself: while the human contribution provided the original conceptual breakthroughs and paradigm-challenging insights (like recognizing the "misattribution" and "technological progress as religion"), the rapid development, refinement, and articulation of these ideas derived predominantly from AI capabilities. This distribution of capabilities—human conceptual innovation paired with AI-driven development and refinement—characterizes our current moment in technological evolution. Yet even this balance is temporary, as AGI systems develop increasingly sophisticated capabilities for original thinking and paradigm challenging.

This framing helps explain why advanced AI systems represent a genuine discontinuity rather than just another step in technological progress. Previous transitions maintained the illusion of human centrality by attributing the productivity of capital to its human operators. The wide proliferation of advanced AI systems breaks this pattern not just by displacing human labor but by making the historical misattribution of productivity unsustainable. The economic implications are profound, as this completes the shift of all productive value to the owners of advanced AI systems, leaving human labor not just less valued, but ultimately irrelevant to the productive process.

The misattribution thesis thus provides both a clearer understanding of historical technological change and a more accurate framework for anticipating the economic wealth inequalities implications of Advanced AI Systems. It suggests that conventional approaches to technological unemployment and reemployment are fundamentally misguided because they fail to recognize how productivity gains have actually functioned throughout industrial history. This understanding is crucial for developing appropriate responses to the challenges Advanced AI Systems present.

Technological Progress As Religion

Notably, when initially presented with the thesis that historical technological unemployment and reemployment experiences will not apply to Advanced AI Systems, three leading AI systems—Perplexity, Deepseek, and ChatGPT—each defaulted to defending the applicability of traditional patterns of technological unemployment and reemployment. Their responses reflexively cited historical examples of how technological disruption eventually created new jobs, suggesting that Advanced AI Systems would follow the same pattern. This automatic defense of technological continuity reveals how deeply embedded the notion of perpetual technological progress and adaptation has become in contemporary thought. Yet this ingrained optimism may blind us to fundamental discontinuities.

Current literature typically claims several weaknesses in the thesis that advanced AI systems break historical patterns of technological employment. These counter-arguments assert that the analysis underestimates human adaptability and creativity in finding new economic niches, that it fails to account for currently unimaginable future jobs, and that it oversimplifies the supposedly synergistic relationship between human skill and technological capability. Yet these very counter-arguments demonstrate how deeply embedded quasi-religious faith in technological continuity has become in contemporary thought.

The appeal to "human adaptability" represents circular reasoning disguised as analysis: humans will adapt because humans have always adapted. This article of faith ignores Advanced AI Systems' defining characteristic—its ability to fill any new niche itself through its capacity for general problem-solving and continuous self-improvement. Similarly, invoking "roles we cannot yet envision" exemplifies faith-based rather than logical thinking. This document's central insight is that our inability to articulate future human economic roles sufficient to overcome widespread human labor displacement stems not from limited imagination but from Advanced AI Systems' comprehensive capability replacement.

This faith-based thinking particularly manifests in appeals to historical precedent when confronting AI's impacts. Claims that "humans have consistently created new forms of value as technology changed" demonstrate not reasoned analysis but rather an article of faith. This perspective fails to acknowledge that Advanced AI Systems represent a fundamentally different technological paradigm—one capable of replicating both physical and cognitive capabilities across virtually all economic domains. The quasi-religious nature of technological optimism becomes clear when its adherents cannot articulate concrete pathways to widespread meaningful human employment, yet

maintain faith that such pathways will inevitably emerge. This represents belief in salvation through technological progress despite mounting evidence to the contrary.

This pattern of faith-based thinking particularly manifests in attempts to preserve some special role for human contribution in the face of automation. The insistence on finding "collaborative relationships" between humans and machines demonstrates our reflexive need to maintain belief in essential human economic value, even as automation increasingly demonstrates its ability to operate independently. This quasi-religious belief in human indispensability is especially evident in the strategies of professional associations and unions, whose protective measures often rest more on articles of faith about human uniqueness than on sustainable economic arguments. Like many religious beliefs, this faith in technological continuity serves a comforting psychological function while resisting contradictory evidence. It allows us to avoid confronting the possibility that human economic participation might become truly optional rather than merely transformed.

AI's Siren Call

Just as the sirens of mythology lured sailors with promises of transcendent beauty only to lead them to destruction, the promises of AI-accelerated breakthroughs and enhanced human capabilities serve as a modern siren's call that distracts from examining the eventual human costs of AI developments.

In an Advanced AI Systems-driven economy where human labor is largely obsolete, access to the "benefits" AI's developers promise would likely become even more restricted, not less. Consider current trends in pharmaceutical research, where AI systems are already accelerating drug discovery and development. While this promises breakthrough treatments, the economic model still requires massive returns on investment, leading to drug prices that many cannot afford even with current income levels.

The siren's call of AI advancements particularly misleads through its promise of democratized access to expertise. While AI systems might theoretically provide high-quality medical diagnosis or legal advice at lower costs, this ignores how the elimination of these professional career paths would reduce economic mobility and concentrate wealth among Advanced AI Systems owners. The promise of cheaper services becomes meaningless when the means to pay for them has been eliminated.

The counterarguments suggesting that Advanced AI Systems' productivity gains will materially decrease the costs of human sustenance ignore the plight of unemployed individuals without means to acquire basic necessities. In the context of an economy dominated by a few Advanced AI Systems, it may be necessary to drastically alter economic structures and institutions to increase minimal subsistence assistance, provide mandatory equitable access, and secure financing from the economically and politically powerful Advanced AI System owners that control these resources.

The timeline illusion represents another aspect of AI's Siren Call—the comforting belief that even if Advanced AI Systems eventually displace most human labor, this transition will occur gradually enough to allow for societal adaptation. This perspective fails to

recognize two critical realities: first, that unprecedented investments from both private and governmental actors are accelerating AI development at a pace that may overwhelm adaptive capacity; and second, that preparing adequate economic and social systems for widespread labor displacement requires anticipatory action rather than reactive responses. The massive infrastructure required for Universal Subsistence Support cannot be implemented hastily; it demands decades of planning, testing, and gradual implementation. By the time displacement reaches critical levels, it may be too late to develop functioning systems that prevent widespread economic and psychological suffering.

The belief that being seduced by the AI's Siren Call we can consciously steer how we deploy Advanced AI Systems represents perhaps the most dangerous fallacy plaguing current policy discussions. The competitive dynamics between corporations and nations create overwhelming incentives to pursue AI advancement regardless of societal consequences.

As the RFI itself acknowledges, AI development is "critical to fundamental United States interests." This creates a deterministic trajectory where individual actors cannot unilaterally choose restraint without sacrificing competitive advantage. This dynamic functions similarly to an arms race, where the collective result of individual rational decisions leads to outcomes no single actor would choose. The illusion of choice masks what is essentially a forced path, with each technological advancement narrowing future options rather than expanding them. Policy frameworks must acknowledge this reality rather than assuming we retain full agency over how AI systems are developed and deployed across the economy.

Universal Subsistence Support

To mitigate the economic devastation of massive labor displacement, society will need to implement policies similar to a Universal Basic Income to provide economic security. Here we refer to such new assistance to the displaced human workers as "Universal Subsistence Support" to emphasize that such support would be minimal and not earned.

As of April 2024, the annual costs of US Federal assistance (SNAP, TANF, SSI, EITC and housing assistance) for an estimated net 85-105 million beneficiaries were approximately \$261-\$293 billion dollars. The estimated number of beneficiaries represents about 24-30% of the U.S. population, which aligns with general estimates of the proportion of Americans receiving some form of government assistance.

A proposed Universal Subsistence Support (USS), replacing SNAP, TANF, SSI, EITC and possibly parts of housing assistance that would provide a family of four (two adults, two children) an estimated annual support of about \$40,000, would, in broad strokes, increase the costs of Federal assistance from the current \$0.3 trillion dollars to a Universal Subsistence Support of \$7.4 trillion dollars covering nearly everyone in the US. Such a scenario would deny the government income tax revenues from individuals (\$2.2 trillion 2023), payroll taxes (\$1.6 trillion 2023) and fundamentally overwhelm the federal budget.

While it is possible that Advanced AI Systems generated productivity increases may materially diminish the estimated subsistence and healthcare costs, this scenario would nonetheless require a complete reimagining of the economy, taxation, and the role of government. Implementing, managing, and funding such a system would be a massive challenge, likely requiring: heavy taxation on automated production and AI-generated wealth; possibly a form of public ownership of major AI and Aldroids systems; a rethinking of concepts like GDP, productivity, and economic growth; and redefining money itself in an AI-dominated economy.

In this scenario, it's likely that policies shifting away from incentivizing population growth would include: capping child benefits at two children per family; providing free contraception and family planning services; and promoting adoption over biological children. This shift would represent a significant change in social policy and would likely be controversial. It would need to be implemented carefully to avoid discrimination or unintended consequences. Potential population implosion would dramatically reduce the estimated subsistence and healthcare costs of what human population would remain.

This reimagining of the economic system raises profound questions about the nature and purpose of the interests of Advanced AI Systems and the role of government in society. It would require a fundamental shift in economics, ownership, and the social contract.

Economic And Population Downward Spiral

A potential for a severe downward spiral could be triggered by AI-driven labor displacement. The cycle could unfold as follows:

Initial AI/AGI Labor Displacement → Reduced Economic Security/Confidence → Sharp Decline in Birth Rates → Reduced Consumer Base → Further Economic Contraction → Deeper Loss of Economic Confidence → Even Lower Birth Rates → Accelerating Population Decline → Shrinking Markets → More Economic Contraction

This cycle is particularly dangerous because each element reinforces the others. Unlike previous technological transitions where displaced workers could envision alternative employment opportunities for themselves and their potential children, widespread Advanced AI Systems adoption could create a uniquely profound crisis of confidence in human economic value and purpose.

The psychological impact is not just about current economic conditions, but about people's fundamental belief in the possibility of future prosperity and meaningful human contribution. If people lose confidence in their ability to provide a purposeful future for their children beyond mere subsistence, birth rates could fall far more dramatically than current models predict.

This could create a "psychological-economic doom loop" where: loss of economic purpose reduces birth rates; reducing birth rates shrinks the economy; shrinking economy further reduces sense of purpose, leading to even lower birth rates.

This suggests current population decline projections may be too conservative, as they may not fully account for the psychological impact of widespread AI displacement on family planning decisions.

Reimagining Human Purpose

While this document has focused primarily on the economic implications of Advanced AI Systems and potential policy responses like Universal Subsistence Support, we must acknowledge that the challenges ahead may require more fundamental reconsiderations of human purpose and flourishing. Beyond addressing immediate economic displacement, society may need to explore alternative frameworks for meaningful human existence in a post-labor economy. Such frameworks might include intentionally human-scaled communities that preserve and celebrate distinctly human qualities, reconnection with natural environments and rhythms, and social structures that prioritize human relationships over technological efficiency. The unprecedented nature of the technological transition we face may ultimately require us to question our basic assumptions about progress, purpose, and what constitutes a fulfilling human life. While economic policies are necessary to manage the transition, they alone may be insufficient to address the deeper existential and cultural implications of a world where human labor is no longer central to economic production. The development of Advanced AI Systems thus presents not just an economic challenge but an opportunity to reimagine human society itself.

About The Author

Max Abecassis in collaboration with Claude AI (2025-03-01)

Max is an inventor (51 U.S. patents), innovator/entrepreneur (customplay.com), and futurist/philosopher ("Beyond the Romantic Animal" 1970). In collaborations with LLMs, principally Claude, the author has published a sequence of essays at nissim.com investigating the feasibility of establishing one or more "Isolated Self-Sustaining Human Sanctuaries" (ISSHS).